

ROBOTICS: DO YOU HAVE IT IN YOU?

The robotics industry is growing at an exponential pace in India for the last few years. Let's explore whether it could be your forte

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Remember robot Kryten in the famous British TV series *Red Dwarf*? In one of the scenes, he says: "I serve, therefore, I am." Have you ever tried to comprehend the meaning of the statement? Well, it's true. Robotics has grown far beyond the realms of imagination and science fictions. Robots and robotics are more firmly entrenched into our day-to-day life. Though remaining invisible, they serve us in innumerable ways.

If you are a budding electronics professional, ready to accept high-risk challenges, or comfortable with both mechanics and computer programming with a drive to build new things, then robotics could be your forte.

Know the field

Let us first define robotics. It is a multi-disciplinary area, which integrates intelligent control, communications, computer vision, mechatronics, sensor fusion, design and many other aspects on a single platform for better service. Robots help industries to translate their competitiveness and deliver radically enhanced levels of productivity, efficiency and profitability.

According to a CII report, from 2008 onwards the world market for industrial robots is projected to rise by an average of 4.2 per cent per annum, reaching 139,300 units in 2010. In India, the industry is expected to grow at two to two-and-a-half times the global average.

Various players are entering this space. Outsourcing to India is happening in robotics, right from research/in-

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—Raj Singh Rathee, managing director, Kuka India



tellectual property (IP) generation, to converting LISP codes to C codes for industrial robots. The Indian manufacturing arena is positioned to grow bigger in the coming decades. The mobile robotics space expected to receive increasing importance in the coming years include social/interactive, consumer and medical robots. India being an agrarian society, opportunities exist in automating agricultural-related activities.

Raj Singh Rathee, managing director, Kuka India, says: "In the last few years, the automotive original equipment manufacture (OEM) sales have been growing at a good pace. More and more OEMs have started their manufacturing in India. India is now emerging as the new manufacturing hub of the world and one of the top markets for robots."

Where can you be placed?

There are two major spaces in robotics—industrial and mobile service. The industrial robotics is a well-established field where robots are widely used in the manufacturing industry. But the mobile or service robot segment is at the infancy level.

Major robot manufacturing players like ABB, Panasonic, Kuka, Precision Automation and Robotics India

(PARI), Motoman Motherson Robotics, Hi-Tech Robotics Systemz and Fanuc have their presence in India. And start-ups like Gridbot and Robohatah are also gradually making their presence felt in the country. Besides, several small and medium enterprises are also planning to invest in India to enhance their global competitiveness.

"Among the robotics applicants in India, till date, automotive industry is the major player with around 70 per cent stake. Companies like Tata Motors, Hyundai Motors, Mahindra & Mahindra, Ashok Leyland, etc, have dedicated departments for robotics. However, other industries like foundry, plastics, food and beverages, logistics, FMCG, solar, medical and entertainment have also started using robots in a big way," says Rathee.

Service robots may play a crucial role in some industries of the SME sector like forging and diamond, which need precision workmanship.

Robots are also used in the fields of defence, nuclear science, sea exploration, servicing of transmission electric signals, designing of bio-medical equipment, etc.

The day is not too far when robots will find applications in almost every field. From managing commerce to household chores, to security and res-